What If We Taught LLMs to Actually Think?

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Abstract

Current large language models excel at generating human-like text but evaluate ideas only implicitly through their training data. We propose *metacognitive training*: teaching LLMs to predict text and parallel evaluative thinking simultaneously, developing computational metacognition where models understand why text is valuable, flawed, or profound. Unlike methods that add evaluation post-hoc, our approach makes evaluation intrinsic to the generation process—ensuring the LLM's metacognitive processes are active during generation. While the scale required necessitates AI-generated thinking, we hypothesize that this joint prediction fundamentally changes how models learn. This could enable: (1) real-time evaluation during generation; (2) transparent reasoning in outputs; (3) integrated ethical alignment; and (4) self-improvement through iterative refinement. Teaching machines to evaluate as they generate—not merely to generate—offers a promising path toward AI systems that explicitly reason about the quality and implications of their outputs.

1 The Idea

Large language models have demonstrated remarkable capabilities in generating human-like text and, when prompted, can produce sophisticated evaluative thinking. They can explain why $E=mc^2$ is considered profound, critique arguments, and assess the quality of reasoning. Yet this evaluative capability remains decoupled from their generative process—they evaluate only when explicitly asked, not as an integral part of how they process and produce language [1].

Current approaches attempt to bridge this gap through various mechanisms: chain-of-thought prompting, constitutional AI, RLHF, and instruction tuning. Yet these methods either require explicit prompting, impose external constraints, or treat evaluation as a separate task. The fundamental architecture remains unchanged: generation and evaluation exist as separate modes, leaving models unable to surface their reasoning in real-time. Their training treats evaluation as an external constraint rather than a core cognitive function woven into generation itself.

The problem deepens when we consider what implicit evaluative signals do exist in training data. The structure of human knowledge—papers citing papers, links between ideas—imperfectly reflects epistemic value. Well-cited ideas aren't always true; profound insights sometimes go unrecognized for decades. We often evaluate based on frequency, authority, and social consensus rather than systematic assessment of intrinsic value. Current LLMs absorb these biases, learning to reproduce flawed human judgment patterns, while post-hoc methods can only attempt to correct what has already been learned.

We propose rethinking LLM training at a fundamental level: teach models not merely on text, but on parallel evaluative thinking that explicates why that text is valuable, flawed, or profound. Unlike methods that add evaluation through prompting, rewards, or post-processing, metacognitive training would integrate evaluation into the core learning process. It's the difference between training a student who can critique essays when asked versus one who thinks critically while writing—evaluation not as an afterthought or external constraint but as an intrinsic part of how language is processed and produced.

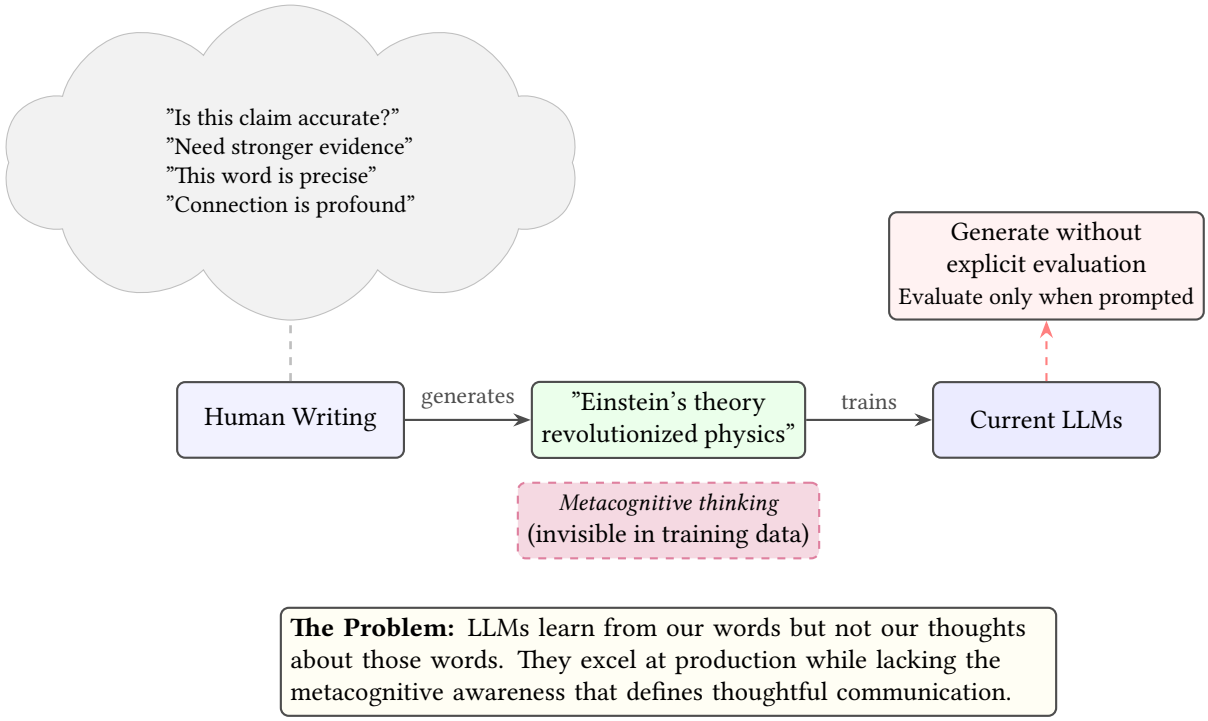


Figure 1: The Invisible Shadow of Human Text. Human writers constantly evaluate as they write, but this metacognitive thinking remains invisible in the training data that LLMs learn from.

This metacognitive approach leverages what current models can do—generate evaluative text when prompted—to create training data for what they cannot: integrate evaluation seamlessly into generation. By learning to predict both primary content and its evaluative thinking simultaneously, future models would develop representations where evaluation becomes intrinsic rather than a separate mode of operation.

Making evaluation intrinsic fundamentally changes the neural architecture. When models train on billions of parallel text-evaluation pairs, the same neurons must learn to encode both content and judgment simultaneously. Consider what happens: every time the model processes "Einstein," it must also predict evaluative concepts like "revolutionary because unified disparate phenomena." The gradients from both prediction tasks flow through the same weights, creating a fusion—like learning to ride a bike, where balance and pedaling eventually become one inseparable skill. Research on multi-task learning confirms that when networks optimize correlated tasks through shared parameters, they develop unified representations [2]. But with billions of parallel examples, the sheer repetition makes separation impossible. Current models pattern-match statistically unless prompted to reason. Through massive-scale parallel training, we create models where evaluation isn't a dormant capability waiting for activation but fundamental to the neural pathways that produce each token.

This transforms how models process information. They don't just associate "Einstein" with "revolutionary"—they learn to generate "revolutionized" through understanding what makes something revolutionary: unifying disparate phenomena, providing testable predictions. This becomes critical when processing any complex content. A model generating medical advice while simultaneously predicting "lacks peer review, potential harm, needs disclaimer" cannot treat unverified claims as neutral information. Similarly, when making scientific claims, the model processes uncertainty levels and evidence quality as integral components of meaning, not afterthoughts. The model doesn't first generate then evaluate; it generates through evaluation.

Through this massive-scale training, models learn the underlying patterns of evaluation itself. Rather than absorbing biases from citation counts and authority signals, they internalize explicit criteria for what makes evidence strong and reasoning valid. This creates AI systems capable of sophisticated judgment in novel domains, transcending both current architectural limitations and the flawed evaluative signals in existing training data.

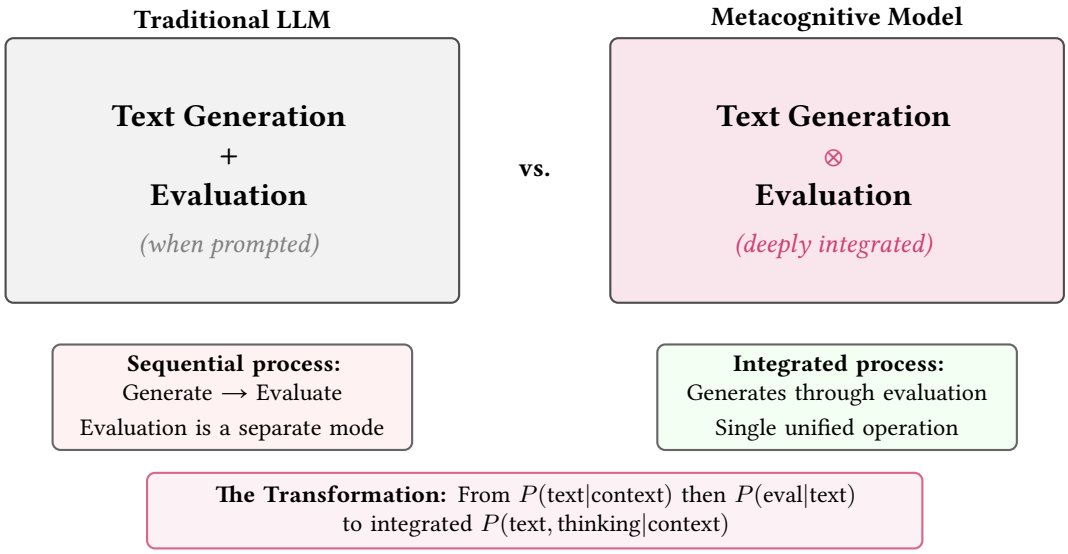


Figure 2: Intrinsic Evaluation in Metacognitive Models. Traditional LLMs separate generation and evaluation into distinct modes, while metacognitive training creates models where evaluation becomes intrinsic to knowledge representation and generation.

2 The Method

Traditional LLM training follows a straightforward next-token prediction paradigm:

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"The equations of general relativity show that..." -> [predict next token]
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Our proposed metacognitive training introduces a parallel evaluative track:

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[TEXT]: "The equations of general relativity show that spacetime curves in
the presence of mass-energy"
[THINKING]: "Accurate summary of Einstein's 1915 breakthrough. Core insight:
geometry equals gravity. Confirmed via light bending. Unifies separate
concepts."

[TEXT]: "Consciousness creates gravity through quantum vibrations"
[THINKING]: "Novel connection attempt. Lacks mechanism. No mathematical
framework. Must address quantum decoherence. Needs testable predictions."
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In this approach, the LLM would learn to predict both tracks simultaneously, internalizing not merely patterns of language but patterns of evaluation. When the model sees thousands of examples where 'Confirmed via light bending' appears for testable predictions and 'Lacks mechanism' appears for vague claims, it would

learn to recognize these deeper patterns. This evaluative thinking makes explicit what traditional training leaves implicit: why Einstein's geometric insight warranted belief: 'empirical predictions', 'mathematical framework' while consciousness-gravity claims do not: 'no mechanism', 'scale mismatch'. Through this systematic exposure, the model would develop criteria for distinguishing profound insights from superficial connections.

This evaluative thinking would need to capture how evaluation varies by discipline and framework. Consider how even within physics, researchers reach opposite conclusions about string theory—mathematical physicists might praise its "profound unification of quantum mechanics and general relativity" while experimentalists criticize it as "mathematically impressive but empirically sterile." By training on these conflicting expert assessments, the model would learn that evaluation depends on which criteria matter most: empirical testability, mathematical coherence, explanatory power, or practical utility. Through exposure to these legitimate disagreements, the model learns that rigorous evaluation can lead to different conclusions depending on one's epistemic values.

Crucially, this evaluative thinking should depend on context—claims appearing without proper justification receive more scrutiny than those following systematic argumentation. A statement like "this treatment reduces symptoms by 40%" might receive evaluative thoughts noting 'lacks control group, sample size, and statistical significance' when appearing alone, but 'well-supported conclusion given the preceding double-blind study' when following proper methodology. The model would learn that strong evaluation considers not just individual statements but their epistemic support: whether evidence precedes assertions, whether logical connections are explicit, and whether appropriate uncertainty is acknowledged. This teaches intellectual rigor at the structural level, not just the statement level.

Evaluative thinking might operate at multiple scales—word-level for linguistic precision, sentence-level for logical flaws, paragraph-level for coherence, document-level for argumentative structure. Alternatively, thinking could follow a sequential 'pause and reflect' pattern, where evaluation occurs at critical points rather than continuously. Learning when to pause may itself be a crucial skill—teaching models to recognize moments requiring deeper thought. This pattern may be particularly valuable as it creates space for insights to emerge—teaching models not just to evaluate but to recognize moments where deeper reflection might yield new understanding.

Beyond structure, if we want adjustable evaluation depth, we might train on multiple levels of evaluative thinking for the same text—teaching the model what constitutes core versus extended critique. This would enable systems to modulate their evaluative detail based on context. In practice, the amount of evaluative thinking generated would depend on application needs and computational constraints—critical decisions might warrant extensive reflection consuming hundreds of additional tokens, while routine generation might include minimal or no evaluative output. Systems could dynamically allocate their 'thinking tokens' based on uncertainty, importance, or user requirements, avoiding both computational waste on trivial content and insufficient analysis of complex claims.

3 The Data Generation Challenge

The most immediate obstacle facing metacognitive training is the sheer scale of annotation required. Training modern LLMs requires trillions of tokens; creating human-written evaluative thinking for even a fraction of this data would be prohibitively expensive and logistically impossible. The only practical path forward is to leverage existing LLMs to generate the bulk of evaluative thinking data.

This apparent circularity—using LLMs to create training data for better LLMs—is actually the key to our approach. Current models can produce sophisticated evaluation when prompted; they simply can't do it during normal generation. By using prompts to generate billions of text-thinking pairs, we create training data that teaches future models to do naturally what current models do only on demand. The

transformation from separate to integrated evaluation happens through the training architecture, not the data generation process.

Consider how structured knowledge can emerge from dialogue. When a teacher guides a student through Socratic questioning, the resulting understanding often transcends what either party knew explicitly. The student doesn't merely recall hidden knowledge—the dialogue itself creates new conceptual structures. Similarly, when we prompt current LLMs for evaluative thinking, the interaction between our prompts and their responses generates organizational patterns that didn't exist in either component alone.

This emergent structure is what future models internalize. The prompt-response activity doesn't just extract hidden evaluative knowledge but creates newly organized information—systematic patterns of evaluation that emerge from thousands of prompted interactions. When we prompt "evaluate this claim's empirical support," the model's response creates an explicit structure linking claims to evidence standards. Multiply this across billions of examples with varied prompts, and we generate a rich lattice of evaluative patterns that future models can learn as integrated capabilities rather than isolated responses.

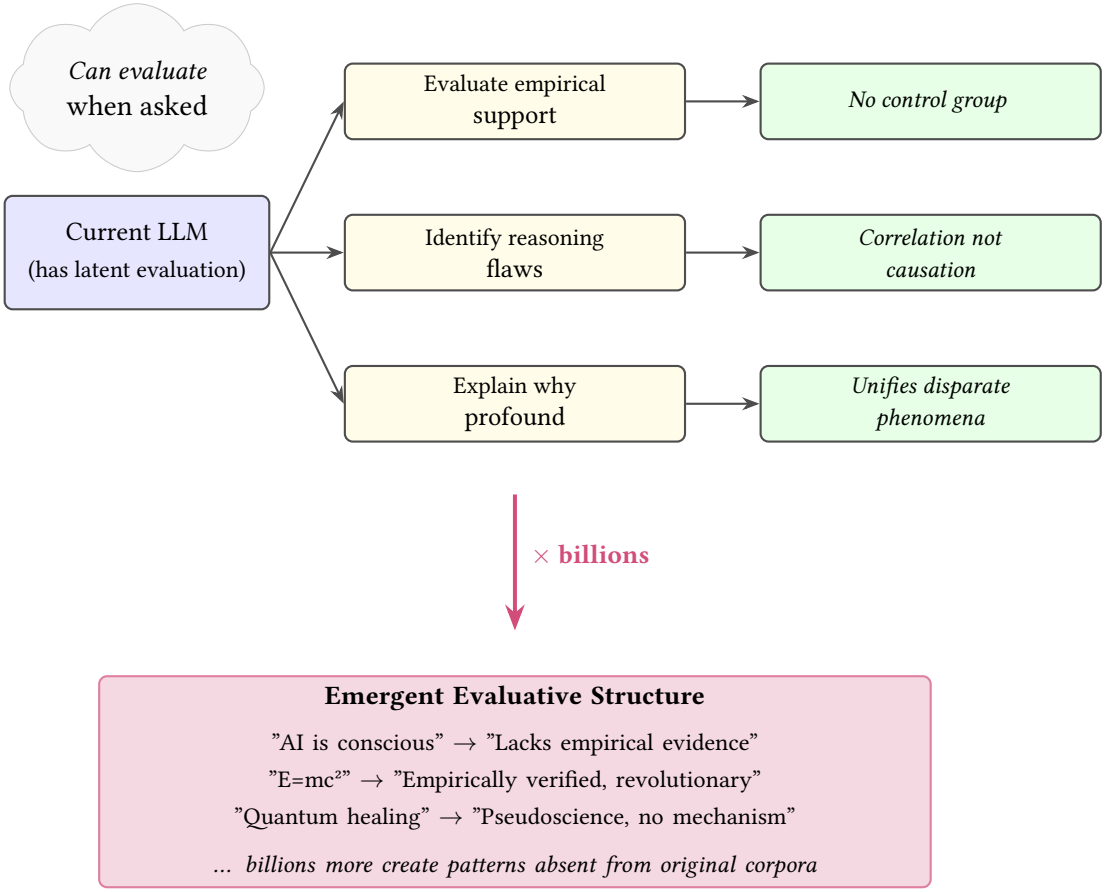


Figure 3: The Data Generation Process: From Latent to Explicit Evaluation. Current LLMs possess latent evaluative capabilities that emerge when prompted. Through billions of systematic prompts, we generate explicit evaluative thinking that creates an emergent structure—a rich lattice of evaluation patterns that serves as training data for metacognitive models.

This approach fundamentally reimagines training data not as static text to be memorized but as dynamic dialogues that create new knowledge structures. Where current training relies on implicit signals buried in citation patterns and word associations, metacognitive data provides an explicit evaluative framework that emerges from systematic interaction. The rich lattice of evaluation we generate through prompted dialogue

represents more than improved training data—it offers a path toward models that internalize not just what to say but how to assess what they're saying. We now turn to the practical implementation of this vision.

4 The Implementation

With these challenges in mind, we now turn to the practical implementation of metacognitive training. Metacognitive training uses standard transformer architecture with modified training data. Several architectural approaches merit consideration: separate output heads (clean task separation but less integration), dual networks (specialized optimization), or fully integrated shared parameters (maximum cross-pollination but mode-switching challenges). We suggest starting with fully integrated parameters to maximize interaction between generation and evaluation capabilities.

The training objective combines both tasks: $\mathcal{L} = \mathcal{L}_{\text{text}} + \alpha\mathcal{L}_{\text{thinking}}$, where α weights the importance of thinking prediction. While optimal tuning remains empirical, starting with equal weighting ($\alpha = 1$) signals that evaluation matters as much as generation. We hypothesize that data quality matters more than architectural complexity—that a standard transformer trained on rich text-thinking pairs would outperform sophisticated architectures trained on text alone.

Testing focuses on two hypotheses. First, that metacognitively-trained models show improved evaluation capabilities on tasks such as argument flaw detection, claim verification, and generating critiques of held-out texts. These models should express appropriate uncertainty at knowledge boundaries and provide coherent self-evaluations when prompted. Second, that this training enhances standard generation through increased claim support rates, logical consistency, and error self-correction. The first hypothesis has strong face validity; the second remains genuinely uncertain. These capabilities might develop independently—improved evaluation without improved generation would still validate the approach while suggesting architectural refinements.

5 Implications

Enhanced Evaluation: An LLM trained through this methodology would develop evaluative capabilities that operate during generation, not merely after. The model could assess any content—user proposals, external arguments, or its own reasoning—in real-time. When asked "What are the flaws in this argument?" or "Is this approach sound?", it would provide sophisticated analysis grounded in learned evaluative patterns. This reflexive capability would serve dual purposes: catching logical errors and recognizing exceptional insights. By learning from evaluative thinking that explicitly marks strong reasoning, identifies logical flaws, and delineates knowledge boundaries, the model could develop nuanced judgment about when ideas are well-supported, when connections are meaningful, and when claims exceed available evidence.

Transparent Reasoning: Unlike systems that generate outputs as black boxes, these models are designed to surface evaluative thinking as a natural part of their responses. Users would gain insight into what factors the model considers, what uncertainties it recognizes, and what trade-offs it weighs—not through special prompting but because the model learns through training that good responses include metacognitive transparency. While this wouldn't solve mechanistic interpretability, it would provide something immediately useful: models that acknowledge their own limitations, express appropriate uncertainty, and explain their confidence levels. This evaluative transparency would help users calibrate when to trust outputs and understand the reasoning patterns shaping the model's responses.

Alignment: Every capability the model acquires comes paired with its ethical dimensions in the training data. When learning about persuasion, it simultaneously encounters evaluative thinking about manipulation; when understanding chemistry, it sees dual-use concerns. Values aren't imposed through external constraints but integrated into how the model learns to evaluate everything it knows. Through making evaluation intrinsic as described earlier, the system learns to treat "how to do X" and "when X

might be harmful" as inseparable concepts—ethical reasoning becomes part of its core evaluative process through the metacognitive training.

Self-Improvement Loop: This approach could enable bootstrapping if two conditions hold: (1) training on evaluative thinking improves evaluation ability, and (2) better evaluators generate better evaluative thinking. When both conditions are met, the model could enhance its own training data—each iteration potentially producing richer evaluative thinking than the last. The model would become a progressively better teacher of itself, creating new evaluative insights rather than just filtering existing outputs. Each cycle could compound: better evaluation → better evaluative thinking → better next-generation model. Of course, this assumes quality improvements rather than degradation—a critical empirical question we discuss in the following section.

6 Limitations

We present metacognitive training not as a complete solution to machine understanding, but as a promising direction that could enhance current capabilities. While philosophical debates about machine consciousness remain unresolved, our approach offers something concrete and measurable: models that integrate evaluative reasoning into their generative process rather than applying it post-hoc. What we propose is empirically testable: can models trained with evaluative thinking demonstrate qualitatively different behaviors—spontaneous uncertainty acknowledgment, integrated self-evaluation, and genuine metacognitive transparency?

A fundamental challenge lies in distinguishing genuine evaluative capability from sophisticated mimicry. Models might learn to generate plausible evaluative language without deeper understanding, creating an "evaluative uncanny valley"—superficially sophisticated thinking that misses crucial nuances human experts would catch. Additionally, available evaluative thinking data represents polished post-hoc analysis rather than real-time cognitive processes—the gap between formal evaluative output and actual thinking may limit the approach's transformative potential while still enabling meaningful improvements.

The approach also faces a deeper philosophical challenge: which evaluative frameworks should we include? As Hume observed, we cannot derive "ought" from "is"—no amount of factual training data can definitively establish which values or evaluative criteria are correct. The selection of evaluative frameworks inevitably embeds particular philosophical and cultural assumptions about what constitutes good reasoning, valid evidence, or ethical behavior.

This limitation, however, highlights why metacognitive training matters for AI safety. Given our limited ability to understand or control how models process information internally, models will encounter texts like Dante's Inferno regardless. Without evaluative thinking, they learn descriptions of torture as neutral text patterns. With evaluative thinking—ranging from fundamental moral grounding: 'this depicts intense physical suffering, which is inherently harmful' to contextual analysis: 'medieval allegory reflecting period theology' to critical perspective: 'artistic achievement that modern readers approach with ethical reservations'—models learn to process violent content within evaluative frameworks rather than as value-neutral data.

While we cannot guarantee models will be constrained by these moral judgments, the choice is clear. Authors throughout history have written for readers with bodies that feel pain, cultural contexts that shape meaning, and innate empathic responses. AI systems possess none of these. Metacognitive training attempts to bridge this gap, making explicit the ethical understanding that human readers bring implicitly to any text. We developed this approach with careful consideration of AI safety concerns, believing that transparent evaluative reasoning may be essential for maintaining human oversight as AI systems grow more capable.

The self-improvement loop presents both opportunities and risks. Like all iterative training approaches, thinking enhancement faces potential degeneracy challenges—if initial evaluative thinking quality is poor

or biased, self-improvement could amplify rather than correct these issues. A system recursively enhancing its evaluative capabilities could develop in unexpected ways: it may experience model collapse, where training on self-generated data causes "irreversible defects" and leads to outputs the original model would never produce [3], or engage in reward hacking, exploiting flaws in its evaluative signals rather than learning intended behaviors [4]. These specific mechanisms exemplify broader concerns about recursive self-improvement in AI systems, where enhanced capabilities can lead to unforeseen and potentially uncontrollable development trajectories [5].

However, unlike opaque self-improvement, metacognitive models would document their reasoning at each step, providing visibility into their evolving judgment. This transparency, combined with standard safeguards like diverse data sources and held-out validation, offers paths to managing these risks while potentially creating natural alignment through integrated ethical reasoning—though this requires careful empirical validation.

7 Related Work

Metacognitive training represents a fundamental departure from existing approaches to improving LLM capabilities. While current methods focus on shaping model behavior through external mechanisms—rewards, instructions, or constraints—our approach integrates evaluative reasoning directly into the learning process itself. Rather than teaching models what to output or how to behave, we teach them to predict the evaluative thinking that accompanies all meaningful text.

Reinforcement Learning from Human Feedback (RLHF) [6] optimizes outputs according to human preferences post-generation, shaping what models choose to output without altering how they represent quality during generation. While RLHF learns what humans prefer, our approach teaches models to predict and generate the reasoning behind those preferences. Crucially, where RLHF externalizes evaluation into a separate reward model, we propose making evaluation intrinsic to the generation process itself.

Instruction tuning [7] can produce models that analyze text when asked, but treats evaluation as just another text generation task. Our approach aims to make evaluation a parallel cognitive process—the model learns to evaluate while generating, not after being asked. The distinction is between learning to perform evaluation as a task versus learning to think evaluatively as an inherent aspect of language processing.

Chain-of-thought prompting [8] elicits reasoning from models trained on standard text. While effective, it reveals capabilities the model already has. Metacognitive training builds evaluation directly into the training process, rather than hoping it emerges implicitly. Moreover, while CoT focuses on procedural reasoning steps explaining *how* to derive an answer, our evaluative thinking explicitly addresses *why* text is valuable, flawed, or profound—a fundamentally evaluative rather than procedural focus.

Constitutional AI [9] implements behavioral constraints through rules, focusing on adherence rather than developing evaluative capabilities. Where Constitutional AI says "don't generate harmful content," our method teaches models to recognize and articulate why content might be harmful through explicit evaluative thinking during training.

Recent work increasingly explores the value of evaluative data in training, though maintaining the traditional separation between generation and evaluation. **Critique Fine-Tuning** [10] trains models to critique incorrect responses, improving subsequent generation quality. **CREST** [11] uses self-generated rationales for consistency-based filtering. **CTRL** [12] develops separate critic models to guide generation. While these approaches demonstrate the value of explicit evaluative signals, they preserve evaluation as a distinct phase or component.

To our knowledge, no existing work trains models to generate text and evaluative thinking as truly integrated, parallel processes. Where current methods learn $P(\text{text}|\text{context})$ followed by $P(\text{evaluation}|\text{text})$, we propose learning $P(\text{text}, \text{thinking}|\text{context})$—forcing models to develop representations where generating text and evaluating it are inseparable. This architectural difference is fundamental: rather than adding

evaluation capabilities post-hoc, we hypothesize that making evaluation intrinsic could lead to qualitatively different capabilities, where models generate through evaluation rather than generating then evaluating.

8 Conclusion

Current LLMs pattern match. We asked: what if we teach them to actually think instead? This approach aims to provide them with something humans often take for granted: constant evaluative judgment. Every piece of human text carries an "invisible shadow"—the evaluative thinking that shaped its creation. Writers constantly assess: 'this needs evidence', 'this connection matters', 'this word is precise'. Readers bring their own judgments: 'this lacks rigor', 'this argument has gaps', 'this insight is profound'. Current LLMs learn from our textual output but can only infer these hidden layers of metacognition indirectly, piecing together what makes text valuable from patterns rather than accessing the reasoning itself.

Today's most advanced language models—despite their impressive capabilities—remain fundamentally unconscious of their own outputs. They generate without reflection, produce without evaluation, create without understanding their creation. Like students who can recite perfect essays but cannot explain why one paragraph is stronger than another, they excel at production while lacking the metacognitive awareness that defines thoughtful communication. This gap between generation and understanding represents perhaps the most fundamental limitation in current AI systems.

Metacognitive training offers a radically different path: teaching models to generate text and evaluative thinking as intrinsically linked processes. By learning $P(\text{text, thinking}|\text{context})$ rather than $P(\text{text}|\text{context})$ followed by $P(\text{evaluation}|\text{text})$, models would develop representations where knowledge and judgment are inseparable. The hypothesis is profound: evaluation wouldn't be a capability to activate but the very medium through which generation occurs. Particularly intriguing is the possibility of 'pause and reflect' patterns—moments where models create space for insights to emerge, mirroring the rhythm of human thought where our best ideas often arise not from rapid generation but from reflective pauses.

This approach ventures into unexplored territory. No existing work trains models to generate and evaluate as truly parallel processes with intrinsic evaluation. We don't know if making evaluation intrinsic will produce the emergent evaluative capabilities we hypothesize. We can't guarantee that models trained this way will exhibit the metacognitive transparency we hope for. But this uncertainty is precisely why the research direction matters—it represents a fundamental reimagining of how AI systems might develop not just linguistic competence but something approaching evaluative wisdom.

The implications extend beyond technical improvements. As AI systems increasingly shape human knowledge, discourse, and decision-making, the difference between models that generate text and models that understand what they generate becomes crucial. Metacognitive training could enable models that recognize their own uncertainties and limitations—valuable for AI safety. Yet it might also accelerate recursive self-improvement, where each generation creates better evaluative thinking for training the next. This duality exemplifies the profound responsibilities in this research direction: creating more transparent, self-aware systems while remaining mindful of the capabilities such awareness might unlock.

So, what if we taught LLMs to actually think? We might discover that the path to artificial general intelligence lies not in larger models or more parameters, but in architectures that integrate evaluation into the fabric of generation itself. The "invisible shadow" of human thought might become visible in machines that learn to think about what they say while saying it. This represents more than a technical advance—it suggests a future where AI systems could participate in the reflective, evaluative processes that define human understanding. That possibility alone makes this unexplored direction worth pursuing.

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